Best Practices for Biomedical Research Data Management

Module 7: Data Sharing and Reuse Data Tools Examples

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This video will give an overview of tools useful across disciplines to manage data in a research workflow and data intensive environments with a slant towards “openly accessible tools”. The tools highlighted are useful to all parts of the research workflow: data hosting services, data manipulation and analysis, data visualization, and publishing.

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- Git repo hosting
- Wikis, issues, social media features
- Distributed Version Control
- Terminal based with available GUIs
- Alternatives - BitBucket, GitLab, etc.

http://swcarpentry.github.io/git-novice/

zenodo

- Cern project - discipline agnostic
- Create curated collections of software, data, papers
- Links with GitHub to allow you to mint DOIs for entire git repos of code releases
- Alternatives - FigShare, Dataverse, etc.

https://guides.github.com/activities/citable-code/

- Originally "IPython Notebooks" but now useful for many languages
  - R, Julia, Python, Ruby and expanding
- Capture full workflows and run your code in a browser interface
- NBviewer - easy rendering of notebooks on GitHub
- Similar features for R in RMarkdown via R Studio

http://jupyter.readthedocs.io/en/latest/index.html

binder

- Turn a GitHub repo into a collection of interactive notebooks
- Run your citable code in a containerized environment with all needed dependencies to reproduce work
  - Click a button to launch your workflow
- Example: LIGO Gravitational waves-

https://losc.ligo.org/s/events/GW150914/GW150914_tutorial.html

http://mybinder.org/

Remember and Encourage Licenses!

http://choosealicense.com/

Publishing/Proposing/Planning

Collaborative, browser-based LaTeX and Markdown editors with Version control

Authorea

https://www.authorea.com/

Overleaf

https://www.overleaf.com/

(md) R

http://rmarkdown.rstudio.com/

Analysis/Data manipulation

- Open source programming languages
- "libraries" and "packages" of useful functions
- integrated development environment is a plus
- Regular Expressions - string manipulation and data restructuring

Analysis/Data manipulation cont.

- Non-programming tools available too
- Proprietary formats, platforms, and cost should be considered

OPEN Refine

ESRI ArcGIS

MATLAB

NVIVO

SPSS

Visualization

- Tools already mentioned facilitate data visualization
- Other tools integrate well with them though via APIs

vega

https://vega.github.io/vega/

plotly

https://plot.ly/

https://d3js.org/

Leaflet

http://leafletjs.com/

Visualization cont.

- Non-programming tools too

tableau public

https://public.tableau.com/s/

Gephi

https://gephi.org/

chartblocks

http://www.chartblocks.com/en/

Bring it all together

https://osf.io/

https://www.globus.org/